Introduction to Artificial Intelligence (AI)

AI trends and applications

Salymbekov University
Miss Aliia Beishenalieva
aliya.beiwenalieva@gmail.com

Recap

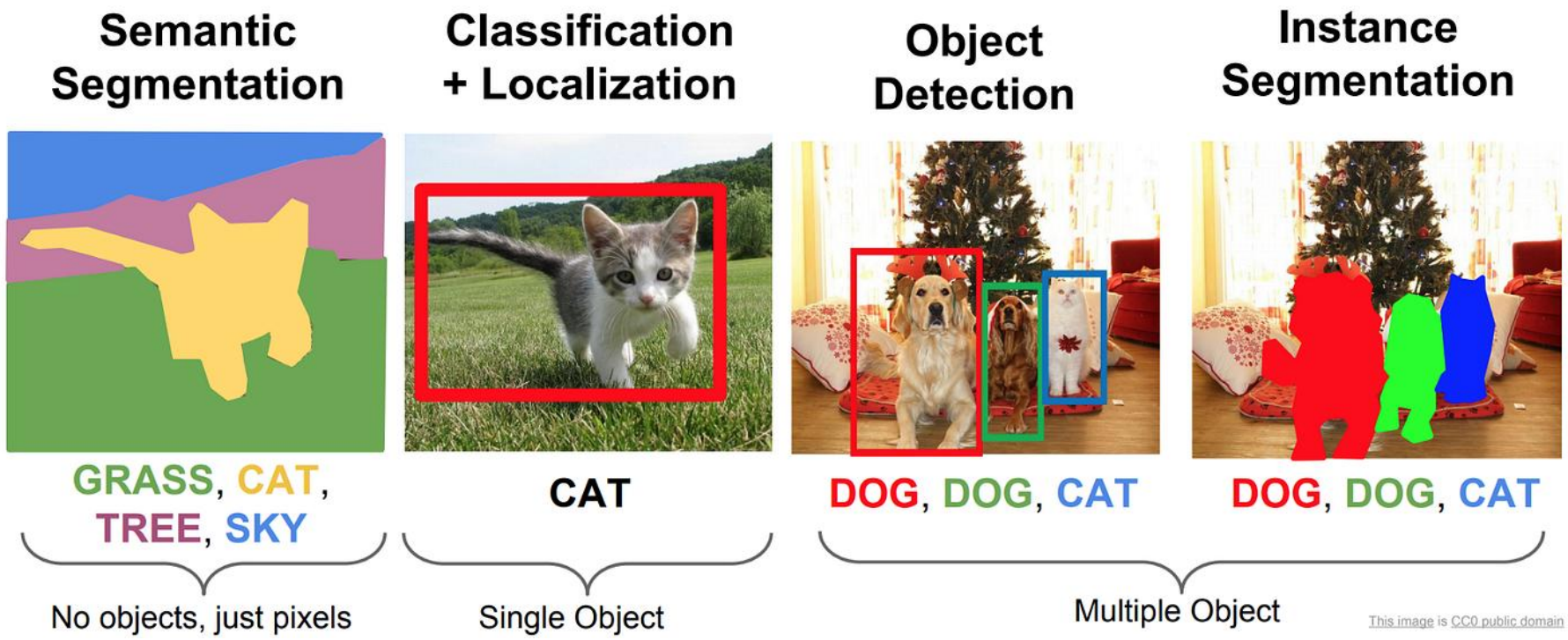
AI Trends

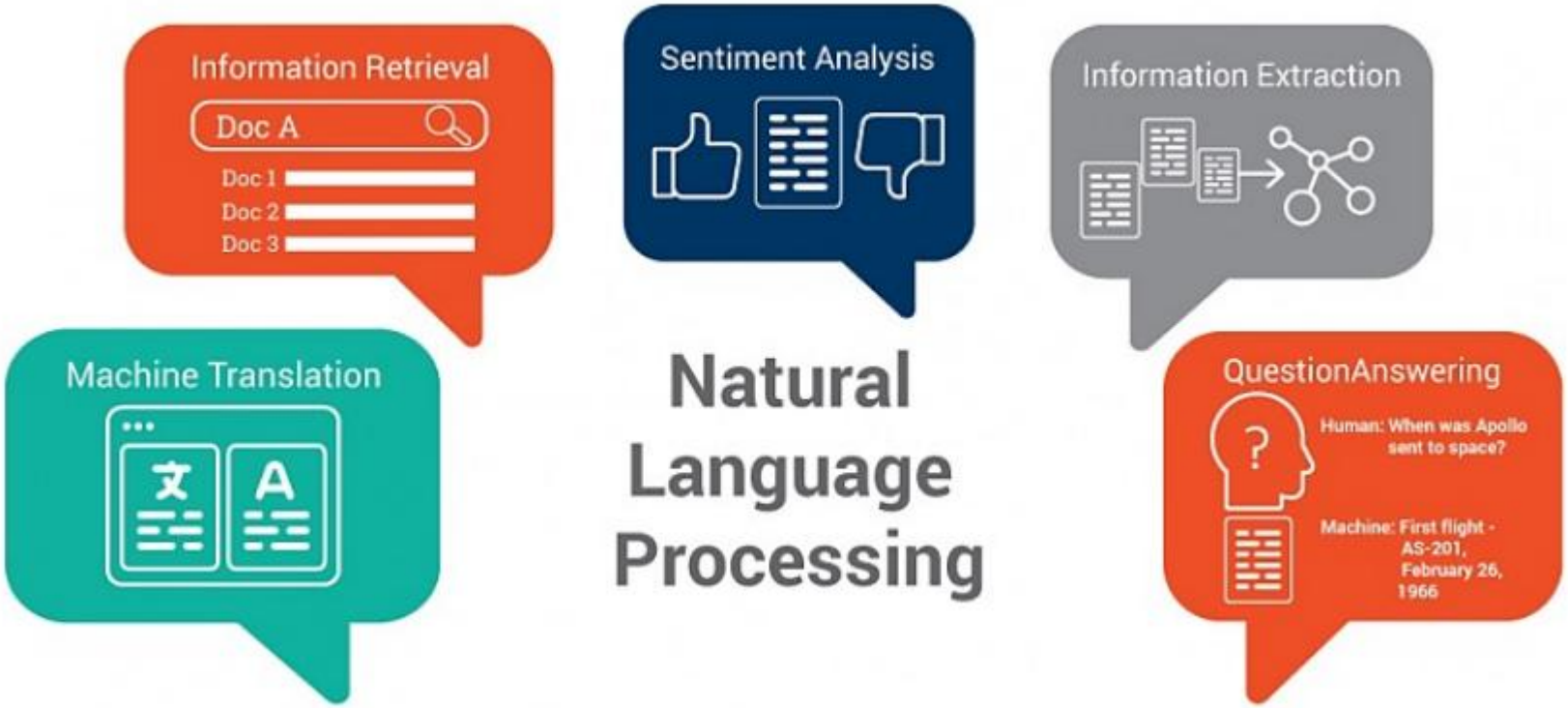
- Computer vision
- Natural language processing
- Generative AI
- AI for Edge Computing

DreamFusion: Text-to-3D using 2D Diffusion (dreamfusion3d.github.io)

https://sites.google.com/view/multinet-inha/demo-videos

Computer Vision



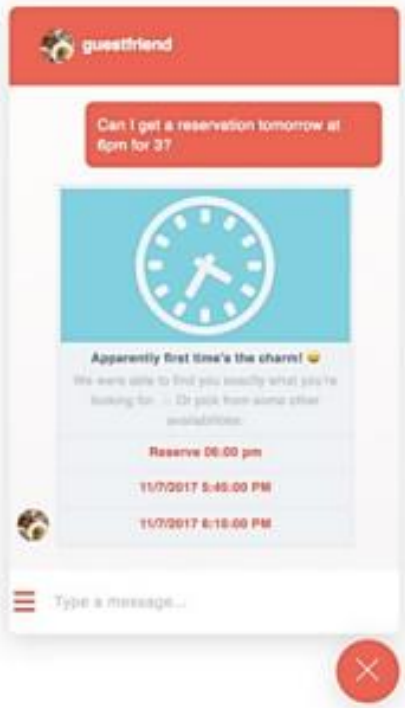

Information Retrieval
Doc A
Doc 1
Doc 2
Doc 3
Sentiment Analysis
Information Extraction
Machine Translation
文
A
Natural Language Processing
QuestionAnswering
Human: When was Apollo sent to space?
Machine: First flight - AS-201, February 26, 1966

Natural Language Processing

- Speech to text (speech recognition)
- Translators (google translator)
- Text classification (spam)
- Chatbots

Chat Bot

WEBSITE

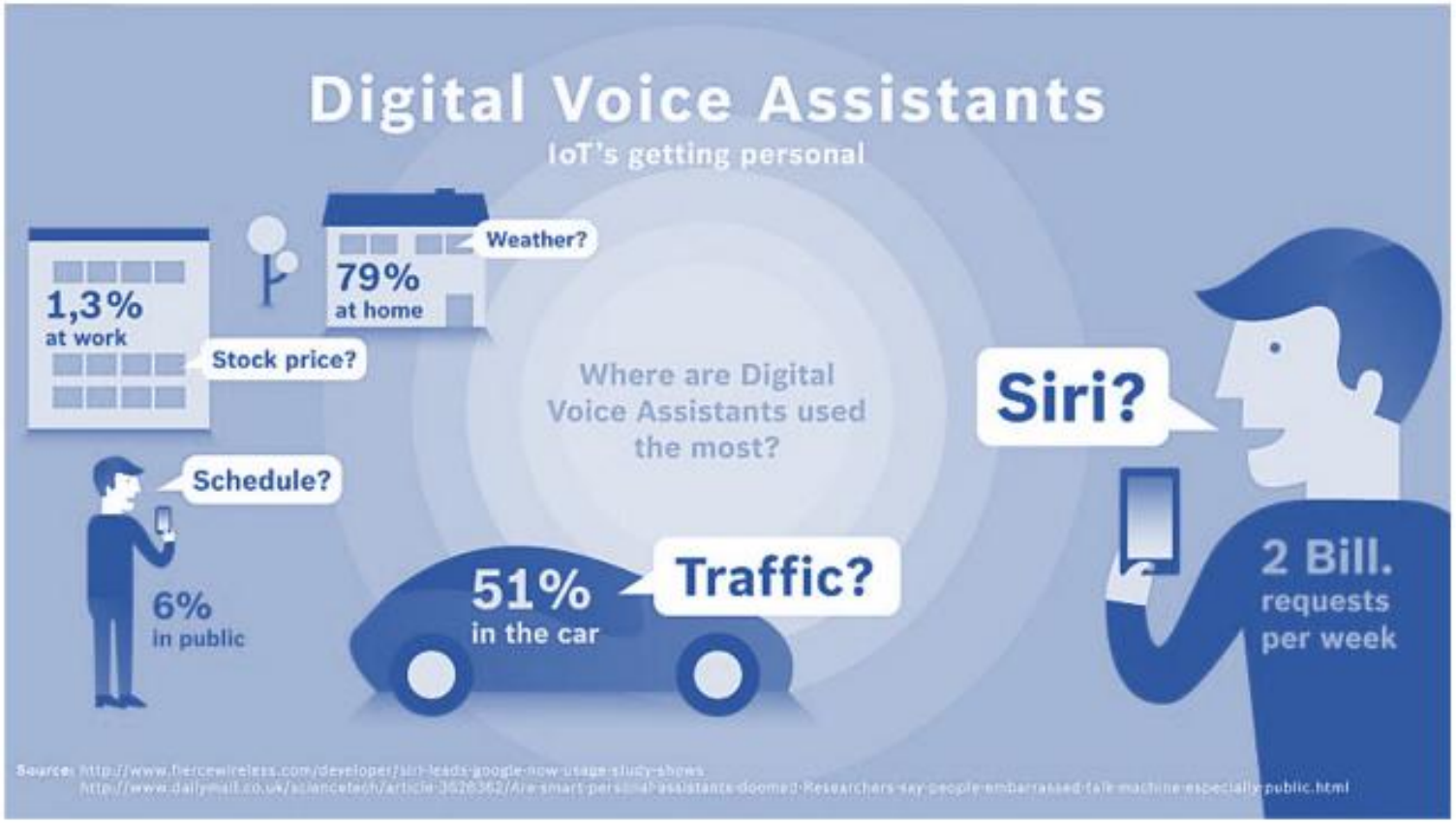


FACEBOOK

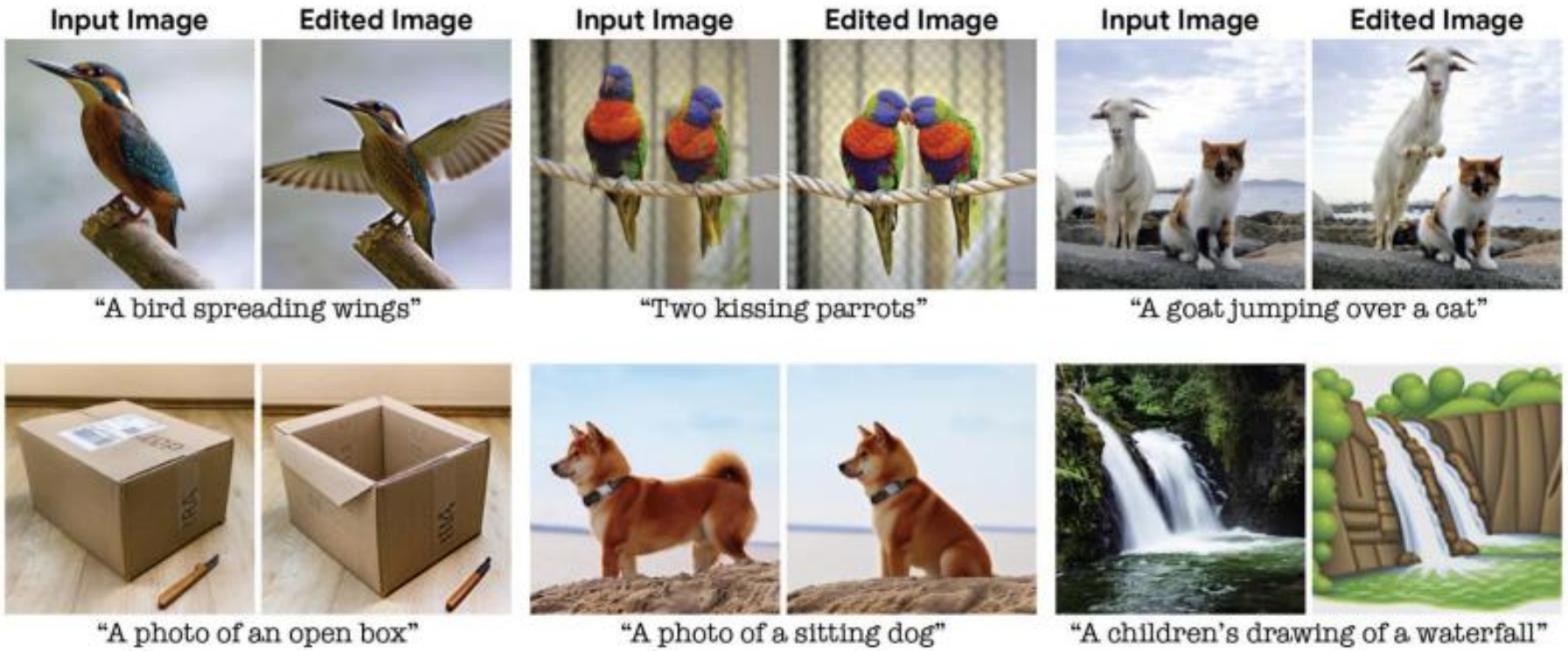
TEXT/SMS

Correction & Assistance

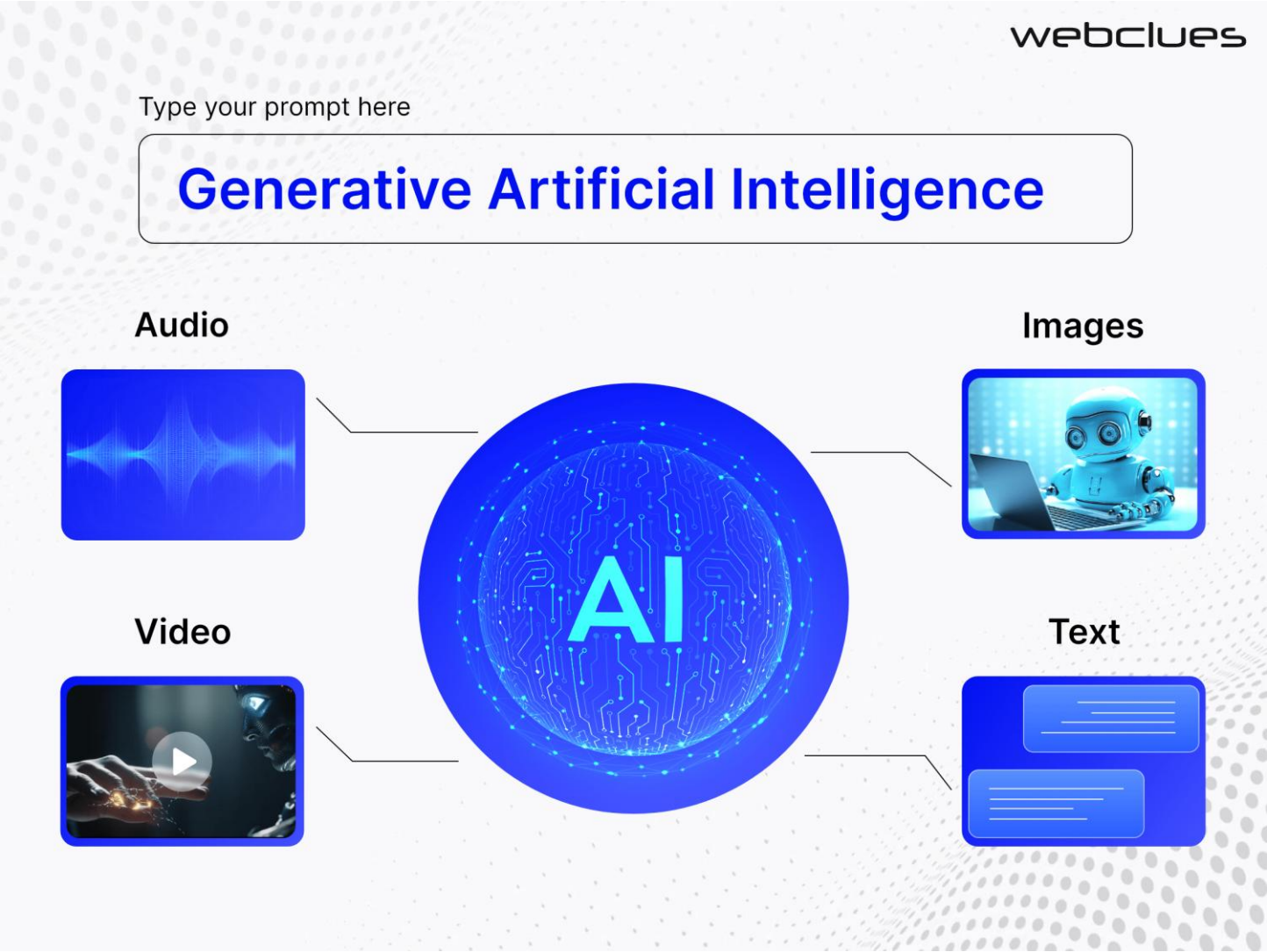
Smart Speaker - Voice Assistant



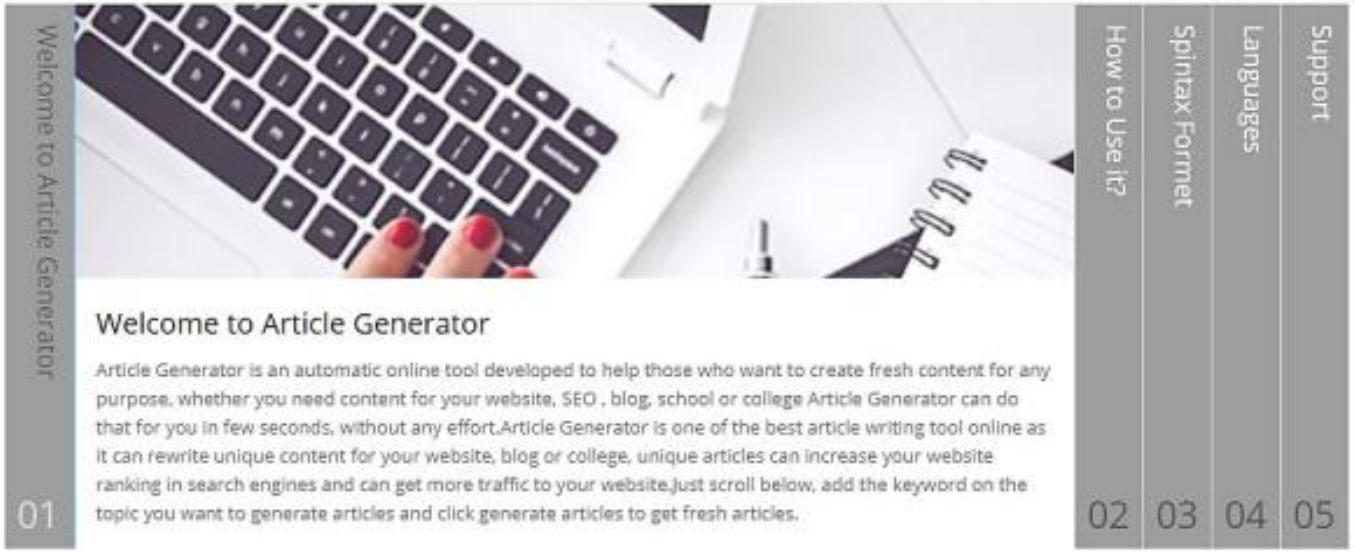
Generative Models



Progress of Generative Models



Article Generation



Keyword

Chell

Search Mode

Public

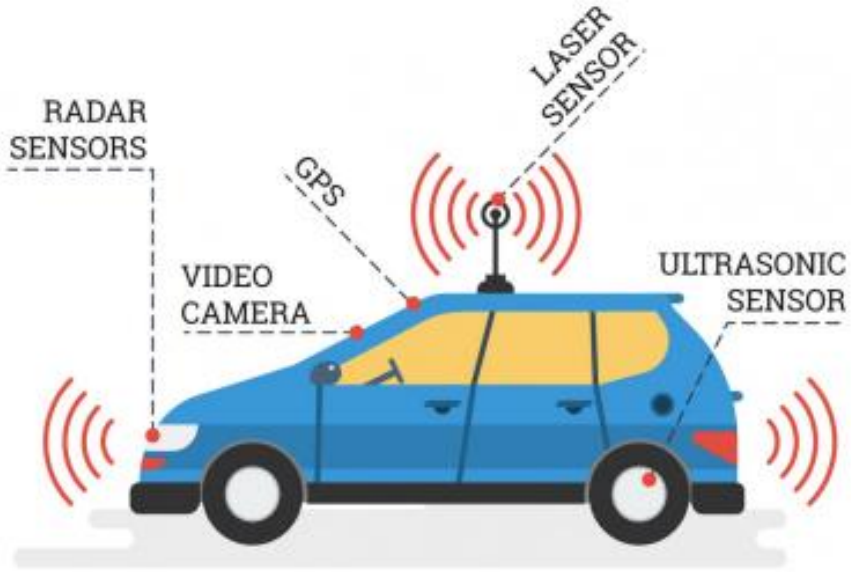
What Language:

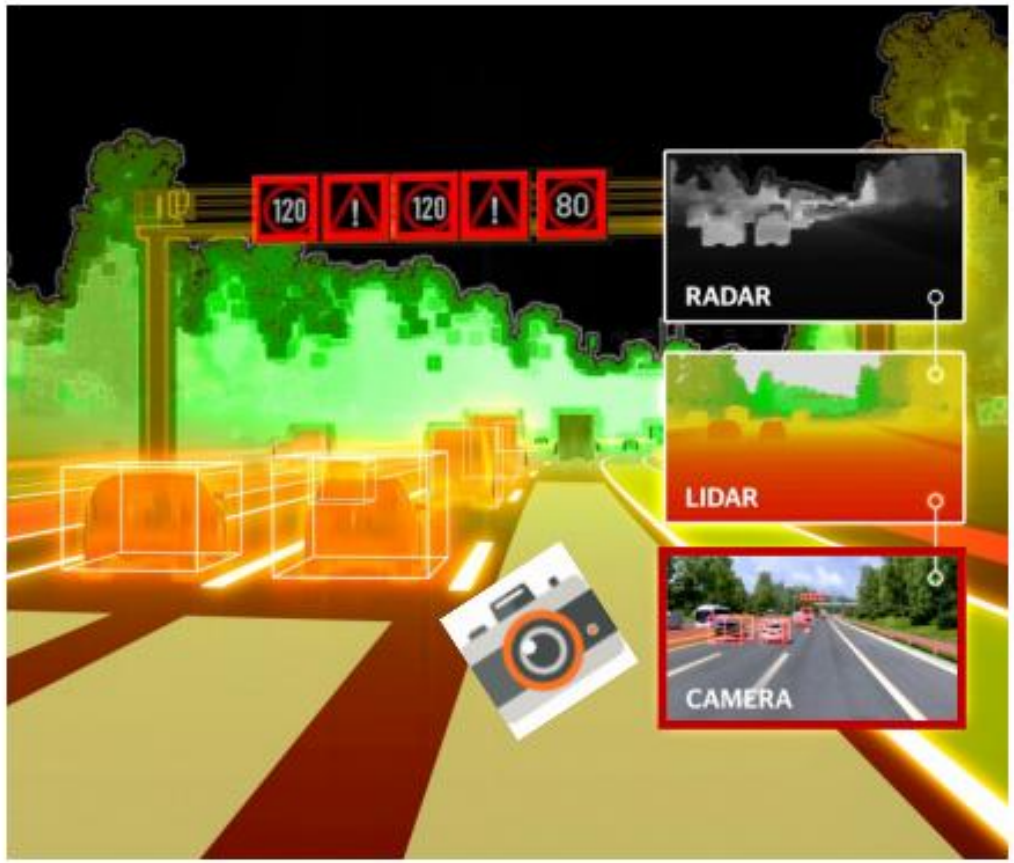
Korean

Voice Generation

NEOSAPIENCE

Multi-Sensor Fusion

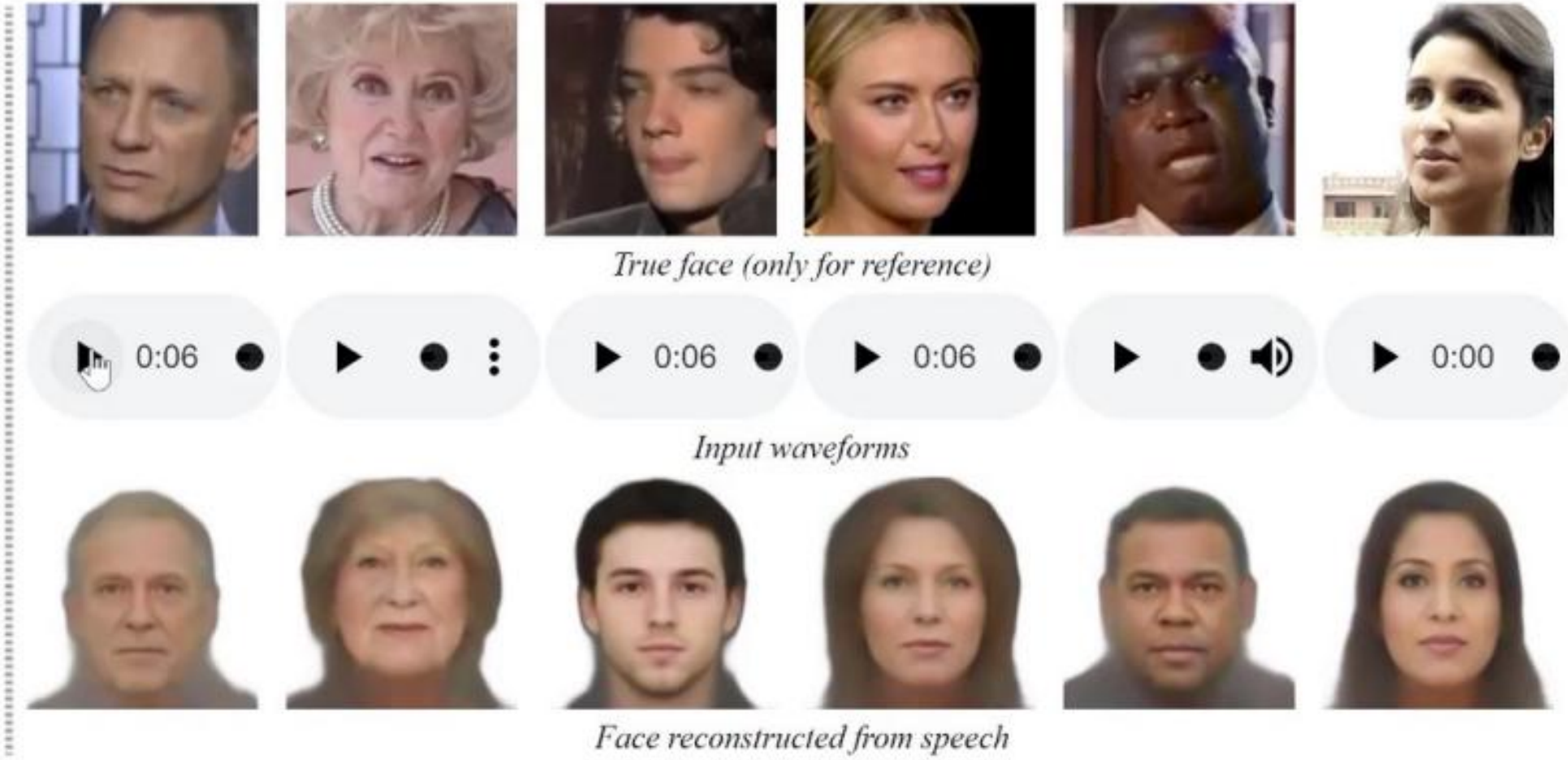




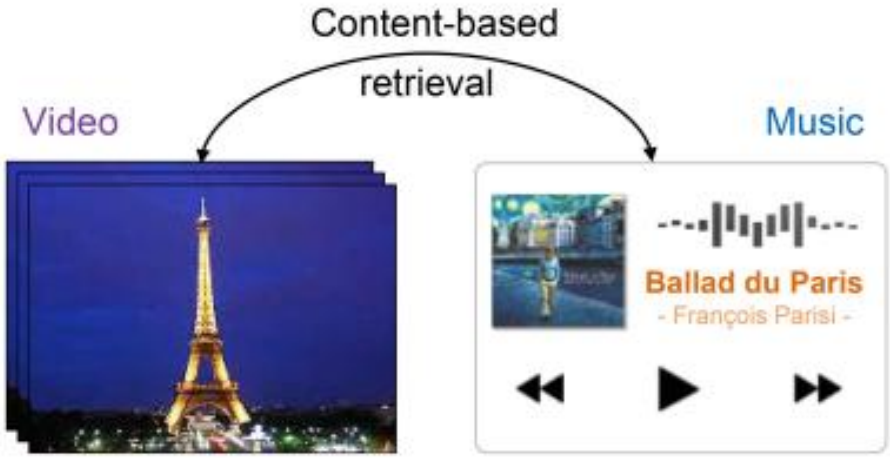
Lip Reading

BBC One HD
07-Apr-2016 22:51:37
THE GOVERNMENT WILL PAY FOR BOTH SIDES

Speech to Face



Video-Music Retrieval



Music video: 205,000 pairs

A sheep that looks like a lion with a pearl earring

진주 귀걸이를 하고 있는 사자 같아 보이는 양

A sheep traveling through time and space

시공을 여행하는 양

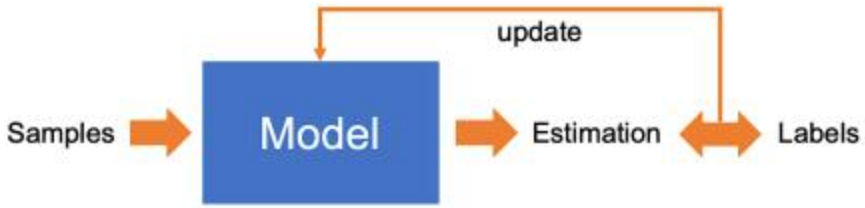
Hacking animals

해킹하는 동물들

Goal/Idea + Data + Model

Data-Driven Approaches

1. Collect a dataset of **samples** and **labels**
2. Use Machine Learning to train a **model**
3. Evaluate the model on new **samples**



What is a Prompt?

What is a deepfake?